A New Method for Calculating Spread in Rod Rolling

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INTRODUCTION

In order to produce high quality rods with high productivity, it is important to develop new techniques of pass design and total control of the rolling process by use of a computer. These techniques, however, cannot be established without mathematical models for predicting precisely the different characteristics of rod rolling. In particular, spread is one of the most important characteristics needed for calculating some other rolling characteristics.

Many reports on spread in rod rolling or caliber rolling have been previously published, and they have proposed some spread formulae from the synthetic evaluation of rolling characteristics.[1-7] But these formulae do not necessarily possess any practical conditions, such as generality, simplicity, and accuracy.

In the present work, the spread phenomena (the profile of the side free surface and the maximum spread) of steel rods in four types of passes were experimentally investigated to develop a new simple formula. These four types of passes included the Square-Oval, Round-Oval, Square-Diamond, and Diamond-Diamond which are illustrated in Table I.

EXPERIMENTAL

A two-high laboratory mill with steel rolls of 200 mm diameter, five calibers or grooves, and roll speed of 14 rpm was employed. The rolling conditions (dimensions of materials and calibers, roll gaps, *etc.*) are included in Table I.

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The materials, mild steel bars about 200 mm long, were heated at 1050 °C in an atmosphere of nitrogen and rolled without lubrication. The widths and heights of the materials were carefully measured at room temperature before and after each pass. The width W_1 can be easily measured from the rolled product with the use of calipers, the color of the contacted area being different from that of the noncontacted area. The contacted area is a little dull compared to the brighter noncontacted area.

RESULTS AND DISCUSSION

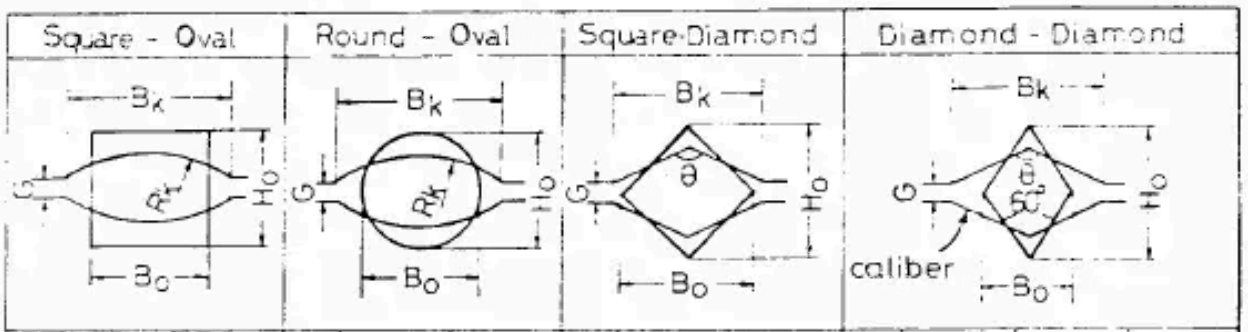
Spread Phenomena

1. *Square-Oval (S-O) pass.* Figure 1 shows the comparison of maximum spread (B_1) and maximum projected width of contact (W_1) after rolling. B_1 is generally larger than W_1 because of the bulging of the side surface. But the difference between them is very small.

2. *Round-Oval (R-O) pass.* Figure 2 shows the relation between the radius ratio (r_1/r_o) of material and the width ratio (B_o/B_o). The factors (r_1, r_o, B_o, B_o) are defined in Figure 2. The ratio (r_1/r_o) which is smaller than 1.0 decreases steeply as B_o/B_G increases. Its level is relatively low at the caliber having a slight curvature. This is because of the large bulging of the side free surface of the material after rolling.

3. *Square-Diamond (S-D) pass.* In this pass, the plastic deformation of the material initially begins at the center of caliber width as in the R-O pass, but the profile of the side free surface does not change before and after rolling except for overfilling. The broken lines in Figure 3 portray the relation between the angle ratio (Θ_1/Θ_0) and the width ratio (B_O/B_G). The ratio (Θ_1/Θ_0) is nearly equal to 1.0 unless the

Table 1. Rolling conditions in four types of passes



	S–O₁	S–O₂		R–O₁	R–O₂		S–D₁	S–D₂		D–S	D–D	D–D₂
B_0	22-25	19-28	B_0	22-28	22-28	B_0	27-34	27-34	B_0	21-28	15-24	14-24
H_0	22-25	19-28	H_0	22-28	22-28	H_0	27-34	27-34	H_0	34-45	27-45	24-42
G	0-1	0-3.5	G	0-2	0-2	G	0-3	0-2	G	0-4.5	0	0
B_k	34.4	35.3	B_k	34.4	35.3	B_k	36.7	36.7	B_k	27.6	36.7	36.7
R_k	20	24	R_k	20	24	θ	115	130	θ	90	115	130
Red	15-33	16-47	Red	5-27	13-38	Red	5-27	16-45	Red	9-31	3-37	11-44
×	□	■	×	○	●	×	△	▲	×	◇	◈	◆

B_0 (mm) : Width, H_0 (mm) : Height, G (mm) : Roll gap } of material
B_k (mm) : Width, R_k (mm) : Radius, θ (°) : Apex angle } of caliber(groove)
Red(%) : Fractional reduction of area × : Marks in figures

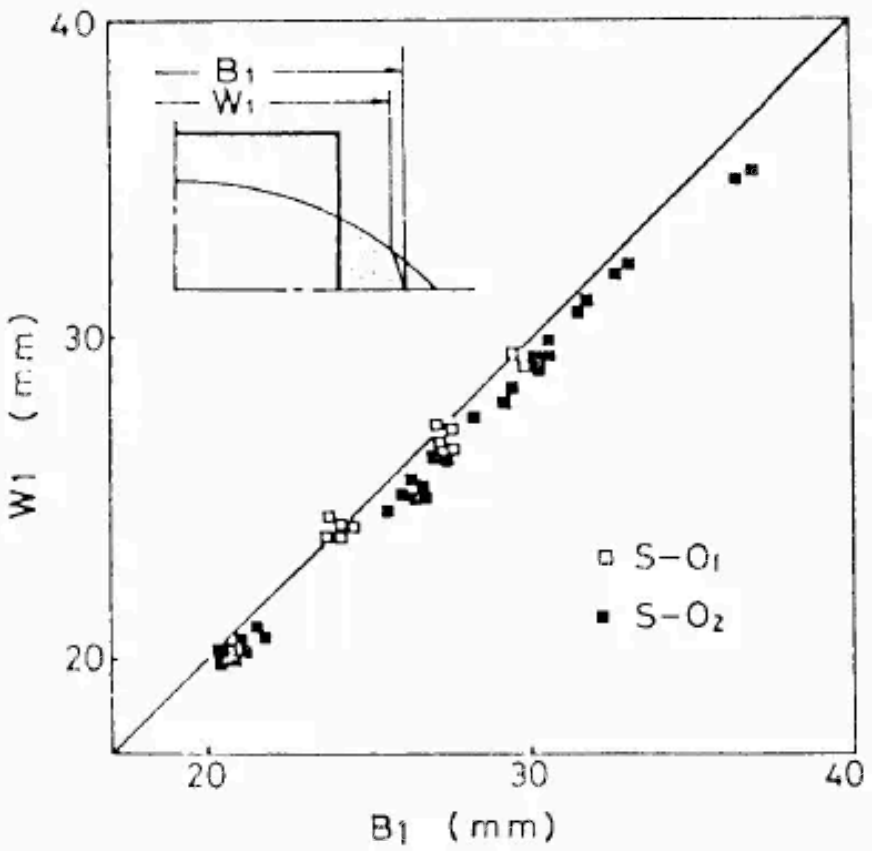


Fig. 1—Comparison of maximum spread (B_1) and maximum projected width of contact (W_1) in Square-Oval passes.

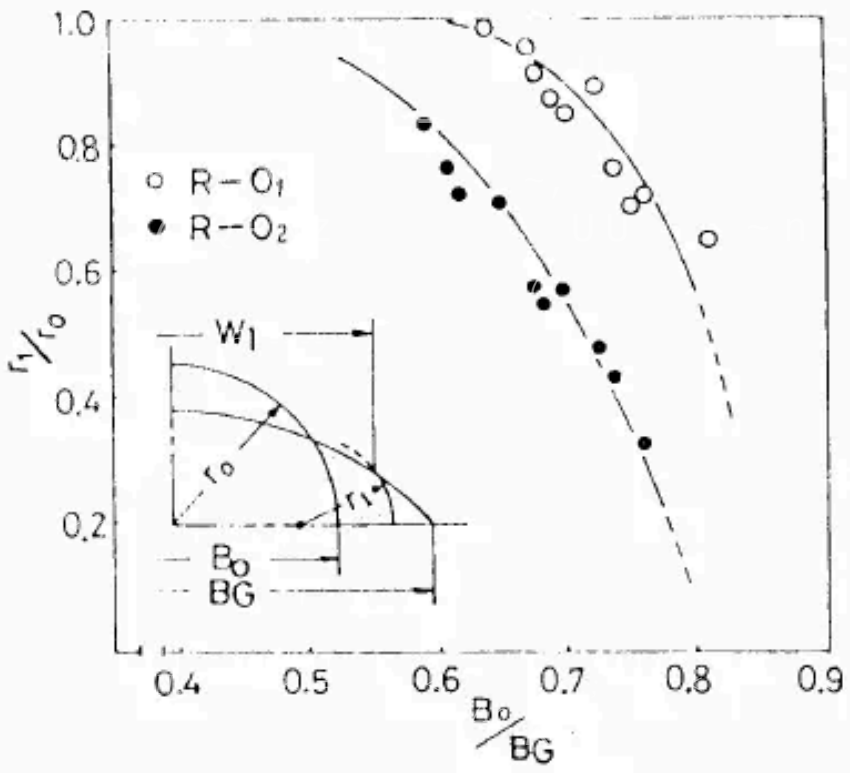


Fig. 2—Relation between radius ratio (r_1/r_0) and width ratio (B_0/B_G) in Round-Oval passes.

product is overfilled in the caliber. In the range of high fillability, the ratio (Θ_1/Θ_0) decreases sharply.

4. *Diamond-Diamond (D-D) pass*. Since the height/width ratio (H_0/B_0) is larger than 1.0, the side free surface does not keep its original profile. As shown by the solid lines in Figure 3, Θ_1/Θ_0 is smaller than 1.0 and decreases gradually with increasing B_0/B_G. This decrease is slight within the range of the small apex-angle of caliber. At the 90° apex-angle, Θ_1 is nearly equal to Θ_0.

Mathematical Models of Side Free Surface

As mentioned above, the profiles of the side free surfaces differ for each type of pass. Figure 4 illustrates the geometrical designations and approximation methods used for describing the profiles of the side free surfaces in the four types of passes. In any pass, the profile is approximated by the straight line or the arc from the above mentioned results.

In the S-O pass, as W_1 is nearly equal to B_1, the profile of the side surface can be approximated by a straight line as

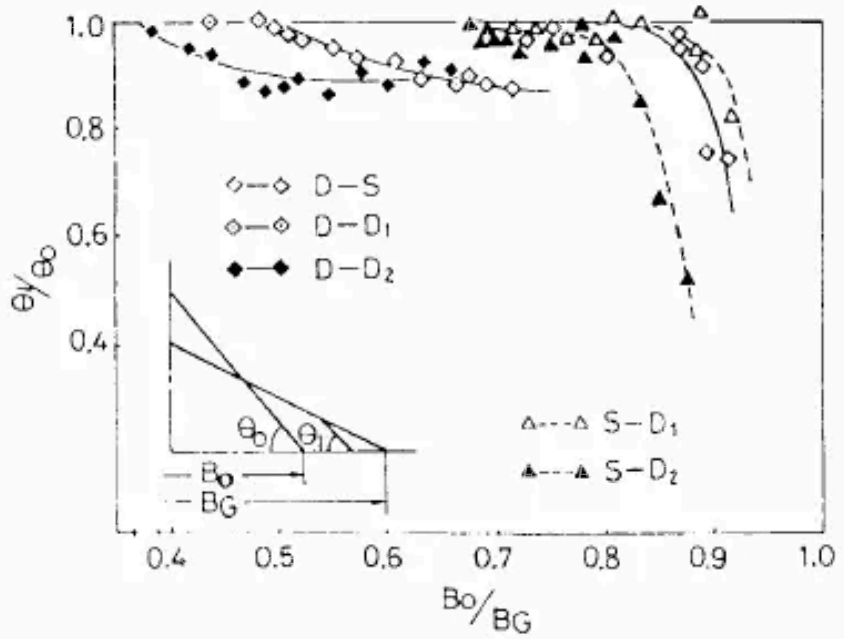


Fig. 3—Relation between angle ratio (Θ_1/Θ_0) and width ratio (B_0/B_G) in Square-Diamond and Diamond-Diamond passes.

illustrated in Figure 4(a). In the R-O pass, the approximation is rather complicated and explained as follows: within the first zone of the coordinates, as shown in Figure 4(b), a circle with its center at the point $(B_1\text{-}B_0)/2$ away from the origin of the ordinate is drawn through the point B_1'. This circular arc gives the exact value of W_1. In the S-D pass, since the right angle of the side surface does not greatly change, the profiles can be regarded as being similar (see Figure 4(c)). The broken straight lines show the extrapolation of the profile after rolling. In the D-D pass, because the angle of the side surface after rolling is smaller than before rolling, as shown in Figure 4(d), the profile is approximated by the straight line connecting the two points B_1' and H_0'.

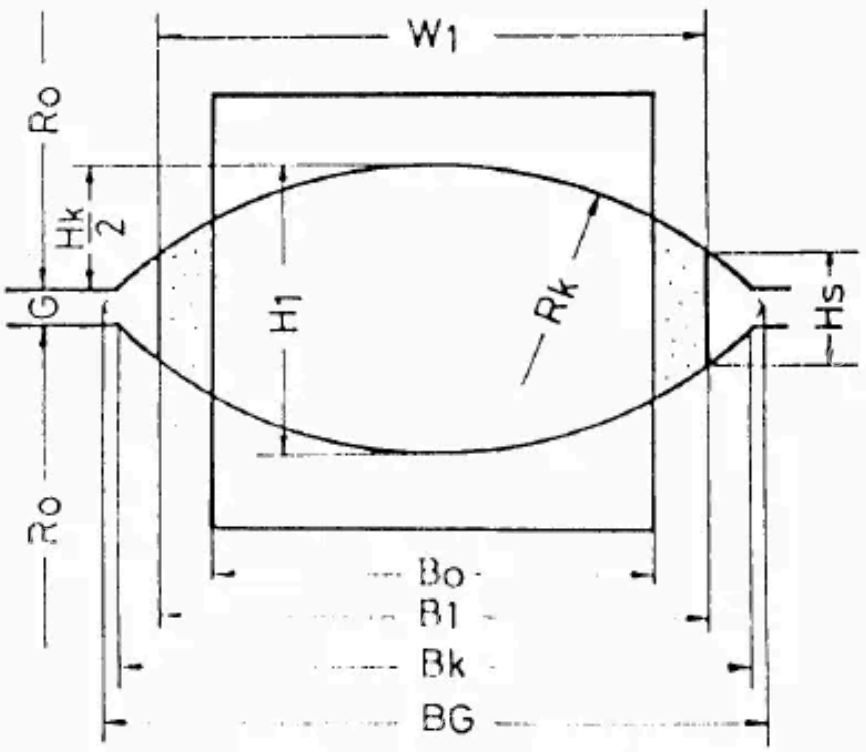


Fig. 4(a)—Geometrical designation and approximation of profile of side free surface in Square-Oval pass.

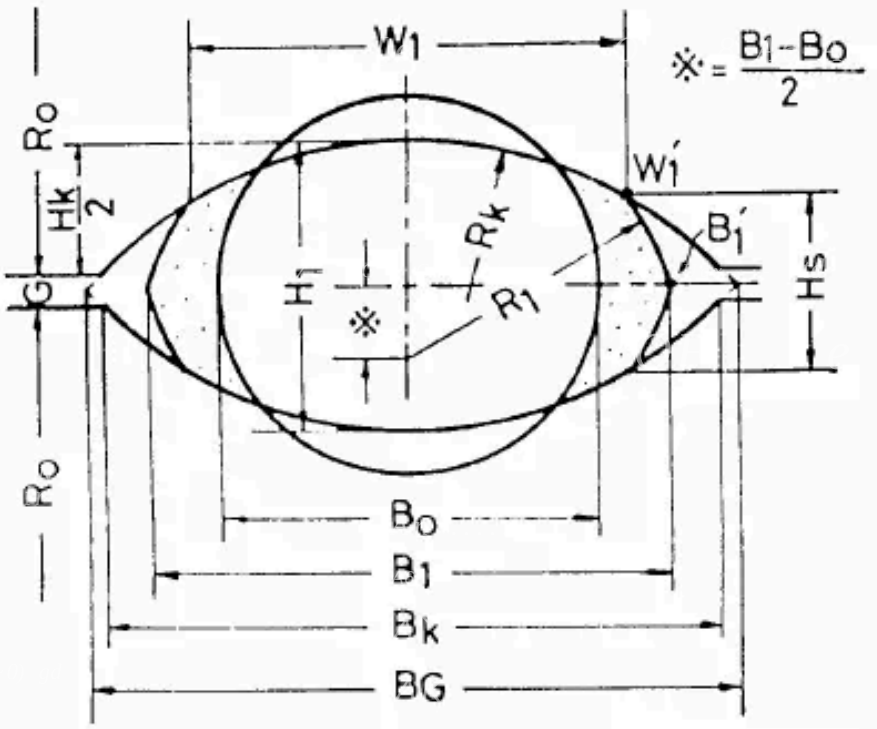


Fig. 4(b)—Geometrical designation and approximation of profile of side free surface in Round-Oval pass.

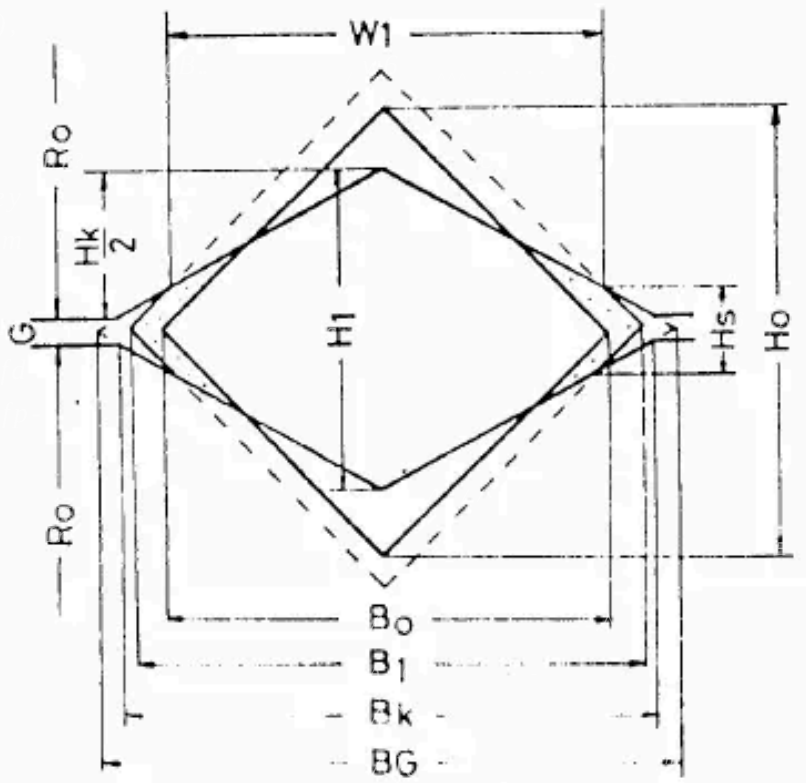


Fig. 4(c)—Geometrical designation and approximation of profile of side free surface in Square-Diamond pass.

According to these geometrical approximations, the profile of side free surface can be described mathematically in each pass and calculated with high accuracy. For example, the formula of each W_1 for the four passes is described simply as follows:

1) in S-O pass

$$W_1 = B_1 \tag{1}$$

2) in R-O pass

$$W_1 = \sqrt{(H_1 - H_s)(4 \cdot Rk + H_s - H_1)} \tag{2}$$

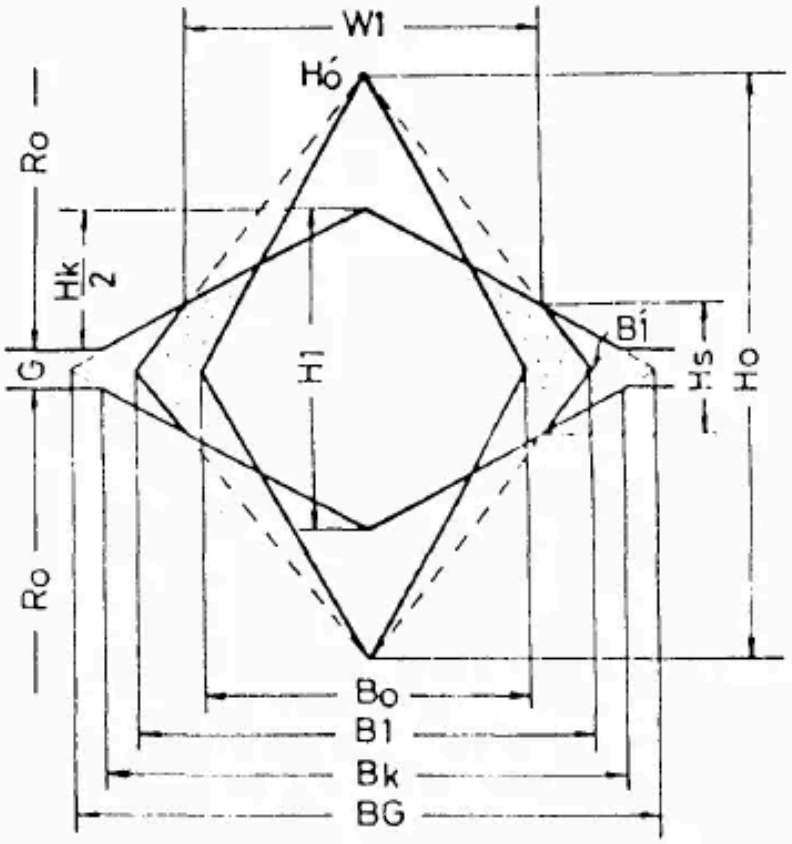


Fig. 4(d)—Geometrical designation and approximation of profile of side free surface in Diamond-Diamond pass.

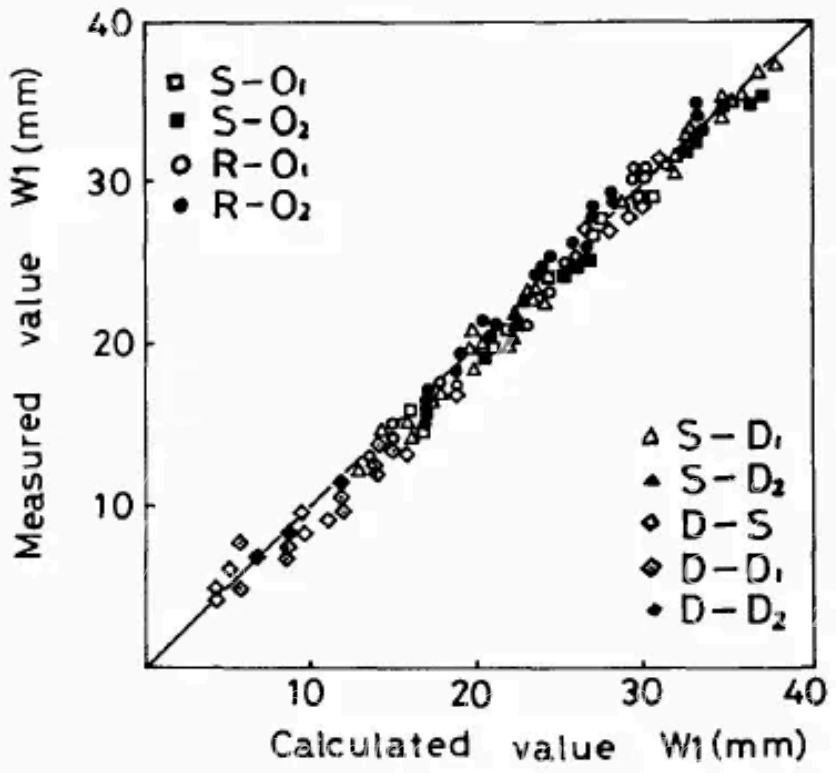


Fig. 5—Comparison of measured and calculated values of maximum projected width of contact (W_1) in Square-Oval, Round-Oval, Square-Diamond and Diamond-Diamond passes.

where

$$Hs = \frac{4(Rk \cdot H_1 - R_1^2) - H_1^2 + (B_1 - B_0)^2}{2(2 \cdot Rk - H_1 - B_1 + B_0)} \quad (3)$$

$$H_1 = Hk + G \quad (4)$$

$$R_1 = \sqrt{\tfrac{1}{2}(B_1^2 - B_1 \cdot B_0 + \tfrac{1}{2} \cdot B_0^2)} \quad (5)$$

3) in S-D pass

$$W_1 = \frac{Bk\,(B_1 - H_1)}{Bk - Hk} \quad (6)$$

4) in D-D pass

$$W_1 = \frac{B_1 \cdot Bk \cdot (H_0 - H_1)}{H_0 \cdot Bk - B_1 \cdot Hk} \quad (7)$$

As shown in Figure 5, measured and calculated values of W_1 for all four passes closely agree.

Moreover, by using these approximations, the cross-sectional areas (F_1) of materials after rolling were calculated and compared with the values measured by a planimeter. A close agreement was obtained between them.

New Experimental Spread Formula

A new spread formula for caliber rolling was developed from a modification of the well-known formulae of flat rolling. Of these formulae, Siebel's formula is very simple and excellent in both generality and accuracy.[8] Hence, this formula was applied to the caliber rolling by using an equivalent rectangular transformation method (Figure 6) and written as

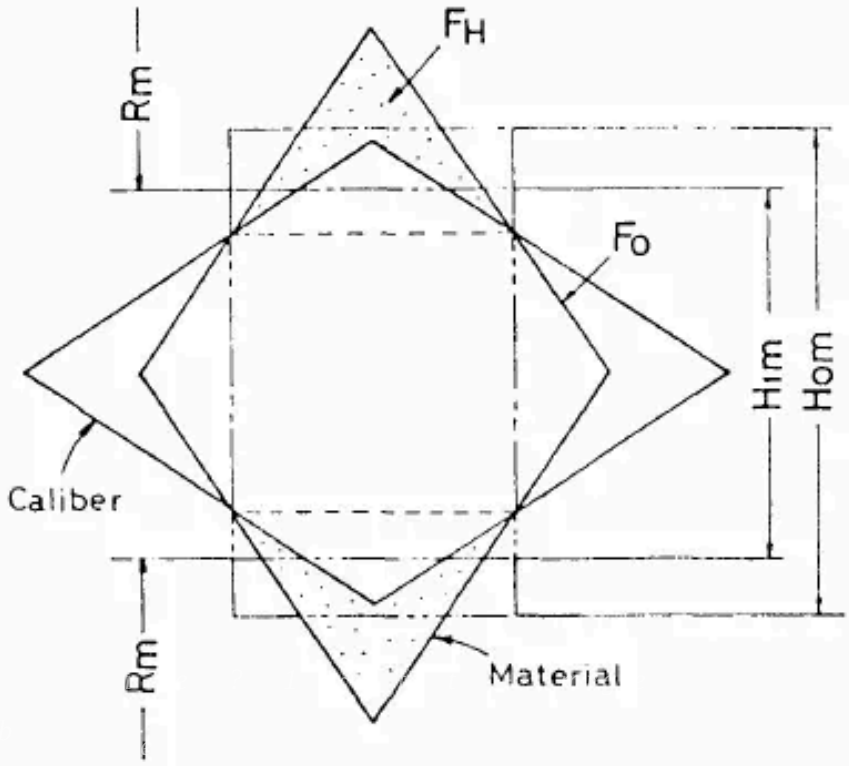


Fig. 6—Equivalent rectangular transformation method to determine mean heights of both material and caliber.

$$\text{Spread: } \Delta B = B_1 - B_0 = \alpha' \cdot Ldm \cdot \frac{H_0m - H_1m}{H_0m} \quad (8)$$

where
- B_0 Width of material before rolling
- B_1 Width of material after rolling
- Ldm Mean projected arc of contact $= \sqrt{Rm\,(H_0m - H_1m)}$
- H_0m Mean height of material before rolling
- H_1m Mean height of caliber
- Rm Mean roll radius
- α' Coefficient.

Although all data of this experiment were applied to (8), the unification among the four passes could not be obtained. This is because the relation between $\Delta B/B_0$ and Ldm/B_0 behaves rather differently in each pass as shown in Figure 7. (On the other hand, the relation between $\Delta B/B_0$ and $(H_0m\text{-}H_1m)/H_0m$ can be unified in all passes).

In order to develop a simple unified formula for all passes, many geometrical parameters were considered and compared with $\Delta B/B_0$. Finally, the following formula was obtained by using the new composite parameter $Y(n)$.

$$\text{Spread ratio: } \frac{\Delta B}{B_0} = \alpha \cdot \frac{Ldm}{B_0 + n \cdot H_0} \cdot \frac{F_H}{F_0} \equiv \alpha \cdot Y(n) \quad (9)$$

where $Y(n)$ Composite parameter at n = unknown
F_H/F_0 Fractional reduction of area (Figure 6)
α Coefficient which is assumed to differ for passes
n Constant for unification.

This formula is equivalent to (8) for $n = 0$. It was proven that the relation between $\Delta B/B_0$ and F_H/F_0 was as well unified as the relation between $\Delta B/B_0$ and $(H_0m\text{-}H_1m)/H_0m$. Therefore, only the relation between $\Delta B/B_0$ and $Ldm/(B_0 + n \cdot H_0)$ was investigated so as to be unified for all four passes. After various values less than 1.0 were substituted in the value of 'n' by trial and error methods. The value $n = 0.5$ was determined to be the best one for unification. The actual data of spread ratio $(\Delta B/B_0)$ are plotted in Figure 8 against $Y(n)$ calculated from the value of $n = 0.5$. The relation between $\Delta B/B_0$ and $Y(0.5)$ is simply unified by the single linear function with error less than ±4 pct for all data of B_1/B_0.

Hence, a new spread formula can be now given as

$$\text{Spread ratio: } \frac{\Delta B}{B_0} = \alpha \cdot \frac{Ldm}{B_0 + 0.5 \cdot H_0} \cdot \frac{F_H}{F_0} \equiv \alpha \cdot Y(0.5) \quad (10)$$

where $Y(0.5)$ is a composite parameter at $n = 0.5$.

The optimum value of α is slightly different for each pass, but its common value for all four passes was found by the least square method to be 0.83.

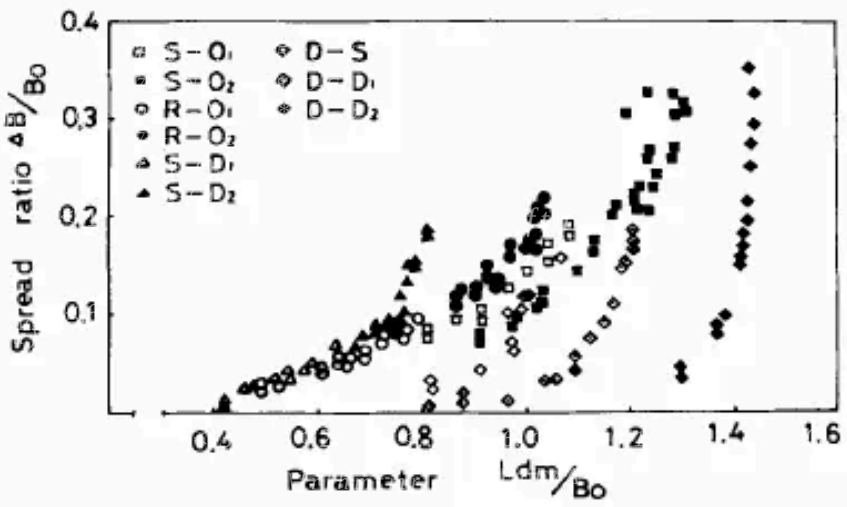


Fig. 7—Relation between spread ratio $(\Delta B/B_0)$ and parameter (Ldm/B_0) in Square-Oval, Round-Oval, Square-Diamond and Diamond-Diamond passes.

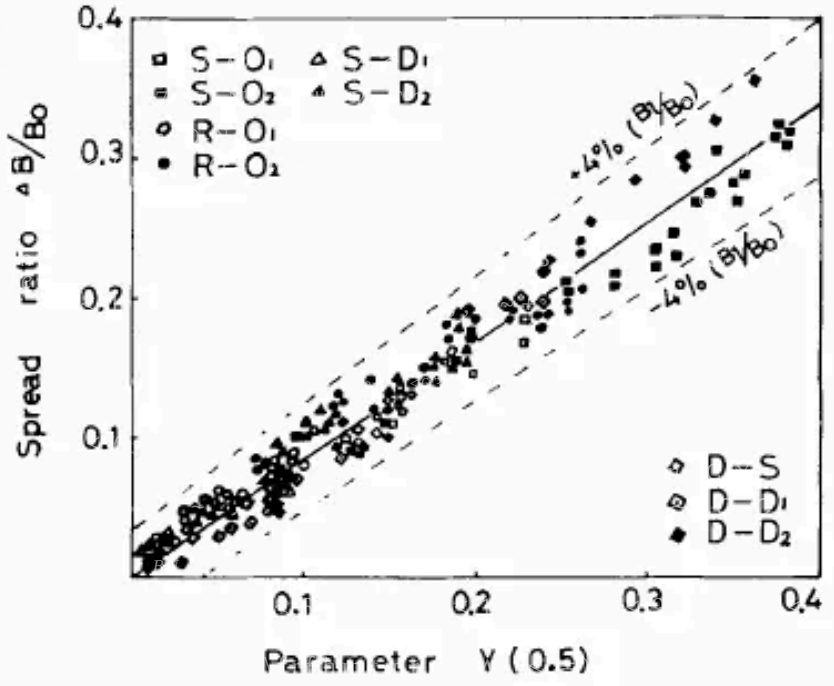


Fig. 8—Relation between spread ratio $(\Delta B/B_0)$ and parameter Y (0.5) in Square-Oval, Round-Oval, Square-Diamond and Diamond-Diamond passes.

APPLICATION OF NEW SPREAD FORMULA

The present work was conducted under restricted conditions, such as roll diameter of 200 mm, five calibers, and roll speed of 14 rpm. In additional experiments carried out by Saito, *et al.*, and authors, it was proven that this new formula can be sufficiently applied in other passes, *e.g.*, Oval-Square, Oval-Round, and Oval-Oval passes, and to the wide range of roll diameter (70 to 550 mm), roll speed (spread appears to be insensitive to this), and temperature (950 to 1150 °C) in both experimental and industrial practice.[9] In these practical conditions, the optimum value α must be selected in the small range of 0.8 to 1.10. Because α is affected by many variables, it is very difficult to determine one value for each pass. However, the values shown in Table II are recommended for practical purposes.

CONCLUSIONS

For the development of a new spread formula for computer control of rod rolling process, the spread characteristics of

Table II. Practical values of α

pass	α
S-O	0.92
R-O	0.97
S-D	0.83
O-S	1.06
O-R	0.83
D-S	0.83
D-D	0.95
O-O	0.95

hot rolling of mild steel rods in the Square-Oval, Round-Oval, Square-Diamond, and Diamond-Diamond passes were experimentally investigated. The conclusions are summarized as follows:

(1) The profile of side free surface differs for the four types of passes: in the Square-Oval pass, it is assumed to be a straight line; in the Round-Oval pass, it is approximated by two special circular arcs; in the Square-Diamond pass, the angle of side surface is similar before and after rolling; in the Diamond-Diamond pass, its angle after rolling is slightly smaller than before rolling.

The maximum projected width of contact (W_1) and the cross sectional area after rolling (F_1) can be precisely calculated.

(2) A new spread formula has been proposed as follows:

$$\frac{\Delta B}{B_0} = \alpha \cdot \frac{Ldm}{B_0 + 0.5 \cdot H_0} \cdot \frac{F_H}{F_0}$$

This formula is very simple and has only one coefficient α but can predict the spread for every type of pass with practical high accuracy. The common value of α for all four passes is 0.83.

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